Mini Project - 2

Objective

Study and analyze the difference between simulated performance and actual execution of MPI applications using performance modeling tools.

Part A: Analysis

- Use the **predictions already generated in Mini Project – 1**.
- Compare:
    - Actual execution time
    - Simulated execution time (from Mini Project – 1 using):
        - SimGrid
        - LogGOPSim
- Both tools are **open-source (links in slides)**, so you can
    - Explore their source code
    - Understand how they model communication and computation
    - Investigate why differences arise in both tools

Part B: Improvement Strategies

Propose and validate strategies to improve prediction accuracy such that the predicted results are as close as possible to the actual execution results.

Output

- Proposed strategies
- Validation showing improvement (before vs after comparison)